Skills Summary

The Derivative as a Limit and an Instantaneous Rate

Use this reference while completing your worksheet or when studying.

Skill 1 — The limit definition at a point

Definition:

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$$

Simplify the quotient **first**. Substituting $h = 0$ before simplifying gives $0/0$, which is not a value; cancelling the h is legal because the limit only uses h near 0, never $h = 0$.

Example: $f(x) = x^2$; find $f'(2)$.
Step 1. The quotient will be $0/0$ at $h = 0$, so the job is to cancel the h algebraically before the limit is taken.
Step 2. $\dfrac{(2+h)^2 - 4}{h} = \dfrac{4h + h^2}{h} = 4 + h$, and $h \to 0$. $\Rightarrow$ $f'(2) = \mathbf{4}$

Try it: $f(x) = x^2$; find $f'(5)$ from the definition.

check: **10**

Skill 2 — From average rate to instantaneous

Rule: Over a window the quotient is an *average* rate; the limit as the window shrinks is the *instantaneous* rate. With only a table of data there is no limit to take, so estimate with the window **centred** on the point — it straddles the instant instead of leaning to one side.

Example: Distance 8 m at $t = 1$ s and 20 m at $t = 3$ s. Estimate the velocity at $t = 2$.
Step 1. No formula means no limit is available, so I use the window centred on $t = 2$ and read its average rate as the estimate.
Step 2. $\dfrac{20-8}{3-1} = \dfrac{12}{2}$. $\Rightarrow$ about **6** metres per second

Try it: Distance 22 m at $t = 2$ s and 50 m at $t = 6$ s. Estimate the velocity at $t = 4$.

check: **7** metres per second

Skill 3 — Reading a derivative as a rate

Rule: $f'(t)$ is not a number attached to the function — it is the rate at which the output changes per unit of input, at one instant. Its **units are output units per input unit**, inherited straight from the difference quotient. The sign says direction: positive means increasing, negative means decreasing.

Example: $V(t) = t^3 - 3t^2$ gallons after t minutes. Find $V'(t)$ and say what it measures.
Step 1. The question asks how fast the volume is changing, which is the derivative, so I differentiate rather than evaluate.
Step 2. Term by term: $3t^2$ from t^3, and $-6t$ from $-3t^2$. $\Rightarrow$ $V'(t) = \mathbf{3t^2 - 6t}$ gallons per minute

Try it: $s(t) = 2t^3 - 5t$ metres after t seconds. Find the velocity function $s'(t)$.

check: $\mathbf{6t^2 - 5}$ metres per second

Skill 4 — Spotting a limit that is a derivative

Rule: If a limit has the shape $\lim_{h\to 0} \frac{(\text{something involving } a+h) - (\text{a constant})}{h}$, read a off the $a+h$ and read f off the operation applied to it; the constant should equal $f(a)$. Then answer with $f'(a)$ instead of grinding the algebra.

Example: $\displaystyle\lim_{h\to 0} \frac{(4+h)^3 - 64}{h}$
Step 1. The quotient matches the definition's shape, so instead of expanding the cube I identify which function and which point it belongs to.
Step 2. $a = 4$ and $f(x) = x^3$ (check: $f(4) = 64$), so the limit is $f'(4)$, and $f'(x) = 3x^2$. $\Rightarrow$ **48**

Try it: $\displaystyle\lim_{h\to 0} \frac{(2+h)^4 - 16}{h}$

check: **32**